

Physics - 3A

Motion in Two Dimension

NEET FOUNDATIONAL WORKSHEET

Name: _____

Roll No. _____

Date: _____

Max. Marks: 140

1. Two bodies are thrown with the same initial velocity at angles θ and $(90^\circ - \theta)$ respectively with the horizontal, then their maximum heights are in the ratio:
- (a) 1 : 1 (b) $\sin \theta : \cos \theta$
(c) $\sin^2 \theta : \cos^2 \theta$ (d) $\cos \theta : \sin \theta$
2. The position vector of a particle is $\vec{r} = (a \cos \omega t)\hat{i} + (a \sin \omega t)\hat{j}$. The velocity vector of the particle is
- (a) directed towards the origin.
(b) directed away from the origin.
(c) parallel to the position vector.
(d) perpendicular to the position vector.
3. For motion in two or three dimensions, the angle between velocity and acceleration is:
- (a) 0°
(b) 90°
(c) 180°
(d) Any angle between 0° and 180°
4. The position of particle is given by $\vec{r} = 2t^2\hat{i} + 3t\hat{j} + 4t\hat{k}$ where t is in second and the coefficients have proper units for $\vec{r}$ to be in metre. The $\vec{a}(t)$ of the particle at $t = 1$ sec is:
- (a) 4 ms^{-2} along y-direction
(b) 3 ms^{-2} along x-direction
(c) 4 ms^{-2} along x-direction
(d) 2 ms^{-2} along z-direction
5. In the projectile motion, if air resistance is ignored, the horizontal motion is at:
- (a) constant acceleration (b) constant velocity
(c) variable acceleration (d) constant retardation
6. The speed of a projectile at its maximum height is $\frac{\sqrt{3}}{2}$ times its initial speed. If the range of the projectile is P times the maximum height attained by it, then P equals:
- (a) $\frac{4}{3}$ (b) $2\sqrt{3}$
(c) $4\sqrt{3}$ (d) $\frac{3}{4}$
7. A car travels due north with a uniform velocity. As the car moves over muddy area, mud sticks to the tyre. The particles of the mud as it leaves the ground are thrown
- (a) vertically upwards. (b) vertically inwards.
(c) towards north. (d) towards south.
8. If the over bridge is concave instead of being convex, the thrust on the road at the lowest position will be:
- (a) $mg + \frac{mv^2}{r}$ (b) $mg - \frac{mv^2}{r}$
(c) $\frac{m^2v^2g}{r}$ (d) $\frac{v^2g}{r}$
9. The ratio of angular speed of a second-hand to the hour-hand of a watch is:
- (a) 3600 : 1 (b) 720 : 1
(c) 72 : 1 (d) 60 : 1

RESPONSE
GRID

1. (a) (b) (c) (d)

2. (a) (b) (c) (d)

3. (a) (b) (c) (d)

4. (a) (b) (c) (d)

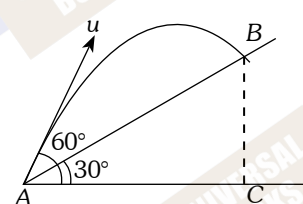
5. (a) (b) (c) (d)

6. (a) (b) (c) (d)

7. (a) (b) (c) (d)

8. (a) (b) (c) (d)

9. (a) (b) (c) (d)

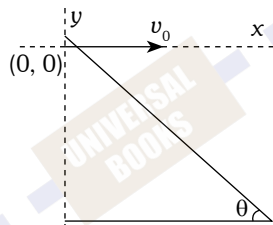
10. A heavy mass is attached to a thin wire and is whirled in a vertical circle. The wire is most likely to break
- when the mass is at the highest point of the circle.
 - when the mass is at the lowest point of the circle.
 - when the wire is horizontal.
 - at an angle of $\cos^{-1}\left(\frac{1}{3}\right)$ from the upward vertical.
11. During a projectile motion if the maximum height equals to the horizontal range, then the angle of projection with the horizontal is:
- $\tan^{-1}(1)$
 - $\tan^{-1}(2)$
 - $\tan^{-1}(3)$
 - $\tan^{-1}(4)$
12. A body is projected at such an angle that the horizontal range is three times the greatest height. The angle of projection is:
- $25^{\circ}8'$
 - $33^{\circ}7'$
 - $42^{\circ}8'$
 - $53^{\circ}8'$
13. The range of a rifle bullet on level ground is 6000 m. The range at an incline of 30° is:
- 4000 m
 - 2000 m
 - 6000 m
 - 1000 m
14. A person can throw a stone to a maximum distance of h metre. The greatest height to which he can throw the stone is:
- h
 - $h/2$
 - $2h$
 - $3h$
15. A projectile can have the same range R for two angles of projection. If t_1 and t_2 be the times of flight in the two cases, then what is the product of two times of flight?
- $t_1 t_2 \propto R^2$
 - $t_1 t_2 \propto R$
 - $t_1 t_2 \propto \frac{1}{R}$
 - $t_1 t_2 \propto \frac{1}{R^2}$
16. The friction of the air causes vertical retardation equal to one-tenth of the acceleration due to gravity (Take $g = 10 \text{ ms}^{-2}$). The time of flight will be decreased by
- 0%
 - 1%
 - 9%
 - 11%
17. A number of bullets are fired in all possible directions with the same initial velocity u . The maximum area of ground covered by bullets is:
- $\pi \left(\frac{u^2}{g}\right)$
 - $\pi \left(\frac{u^2}{2g}\right)^2$
 - $\pi \left(\frac{u}{g}\right)^2$
 - $\pi \left(\frac{u}{2g}\right)^2$
18. Time taken by the projectile to reach from A to B is t . Then the distance AB is equal to:
- 
- $\frac{ut}{\sqrt{3}}$
 - $\frac{\sqrt{3}ut}{2}$
 - $\sqrt{3}ut$
 - $2ut$
19. The driver of a car travelling at 72 km h^{-1} suddenly sees a big rock on the road at a distance of 20 m. What can he do to avoid a collision?
- Apply brakes
 - Turn sharply
 - Follow a zig-zag path
 - Shut the engine
20. If a cyclist doubles his speed while negotiating a curve, how does the tendency to overturn vary?
- Remains unchanged
 - Doubled
 - Halved
 - Quadrupled
21. For angles of projection of a projectile at angles $(45^{\circ} - \theta)$ and $(45^{\circ} + \theta)$, the horizontal range described by the projectile are in the ratio of:
- 2 : 1
 - 1 : 1
 - 2 : 3
 - 1 : 2
22. A ball is projected from the ground at a speed of 10 ms^{-1} making an angle of 30° with the horizontal. Another ball is simultaneously released from a point on the vertical line along the maximum height of the projectile. The initial height of the second ball is: (Take $g = 10 \text{ ms}^{-2}$)
- 6.25 m
 - 2.50 m
 - 3.75 m
 - 5 m
23. A projectile is thrown with a velocity of 10 ms^{-1} at an angle of 60° with horizontal. The interval between the moments when speed is $\sqrt{5g} \text{ m/s}$ is (Take, $g = 10 \text{ ms}^{-2}$).
- 1 s
 - 3 s
 - 2 s
 - 4 s

RESPONSE GRID

10. (a) (b) (c) (d) 11. (a) (b) (c) (d) 12. (a) (b) (c) (d) 13. (a) (b) (c) (d) 14. (a) (b) (c) (d)
 15. (a) (b) (c) (d) 16. (a) (b) (c) (d) 17. (a) (b) (c) (d) 18. (a) (b) (c) (d) 19. (a) (b) (c) (d)
 20. (a) (b) (c) (d) 21. (a) (b) (c) (d) 22. (a) (b) (c) (d) 23. (a) (b) (c) (d)

24. A man standing on a hill top projects a stone horizontally with speed v_0 as shown in figure. Taking the coordinate system as given in the figure. The coordinates of the point where the stone will hit the hill surface are

- (a) $\left(\frac{2v_0^2 \tan \theta}{g}, -\frac{2v_0^2 \tan^2 \theta}{g}\right)$
 (b) $\left(\frac{2v_0^2}{g}, -\frac{2v_0^2 \tan^2 \theta}{g}\right)$
 (c) $\left(\frac{2v_0^2 \tan \theta}{g}, -\frac{2v_0^2}{g}\right)$
 (d) $\left(\frac{2v_0^2 \tan^2 \theta}{g}, -\frac{2v_0^2 \tan \theta}{g}\right)$



25. Two paper screens A and B are separated by a distance of 100m. A bullet pierces A and then B. The hole in B is 10 cm below the hole in A. If the bullet is travelling horizontally at the time of hitting A, then the velocity of the bullet at A is

- (a) 100 ms^{-1} (b) 200 ms^{-1}
 (c) 600 ms^{-1} (d) 700 ms^{-1}

26. A body of mass m is thrown upwards at an angle θ with the horizontal with velocity v . While rising up the velocity of the mass after t seconds will be

- (a) $\sqrt{(v \cos \theta)^2 + (v \sin \theta)^2}$ (b) $\sqrt{(v \cos \theta - v \sin \theta)^2 - gt}$
 (c) $\sqrt{v^2 + g^2 t^2 - (2v \sin \theta)gt}$ (d) $\sqrt{v^2 + g^2 t^2 - (2v \cos \theta)gt}$

27. The equation of projectile is $Y = \sqrt{3}X - \frac{1}{2}gX^2$. The velocity of projection is

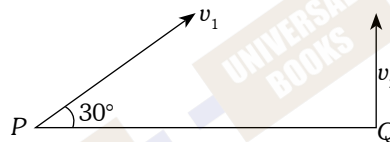
- (a) 1 ms^{-1} (b) 2 ms^{-1}
 (c) 3 ms^{-1} (d) 1.2 ms^{-1}

28. The horizontal range of a projectile fired at an angle of 15° is 50 m. If it is fired with the same speed at an angle of 45° , its range will be

- (a) 60 m (b) 71 m
 (c) 100 m (d) 141 m

29. A projectile A is thrown at an angle 30° to the horizontal from point P. At the same time, another projectile B is thrown with velocity v_2 upwards from the point Q vertically below the highest point A would reach. For B to collide with A the ratio $\frac{v_2}{v_1}$ should be

- (a) $\frac{\sqrt{3}}{2}$
 (b) 2
 (c) $\frac{1}{2}$
 (d) $\frac{2}{\sqrt{3}}$



30. An object is projected with a velocity of 20 ms^{-1} making an angle of 45° with horizontal. The equation for trajectory is $h = Ax - Bx^2$, where h is height, x is horizontal distance. A and B are constants. The ratio $A : B$ is (Take, $g = 10 \text{ ms}^{-2}$)

- (a) 1 : 5 (b) 5 : 1
 (c) 1 : 40 (d) 40 : 1

31. A cart is moving horizontally along a straight line with a constant speed of 30 ms^{-1} . A projectile is to be fired from the moving cart in such a way that it will return to the cart (at the same point on cart) after the cart has moved 80 m. At what velocity (relative to the cart) must be projectile be fired? (Take, $g = 10 \text{ ms}^{-2}$)

- (a) 10 m/s
 (b) $\frac{20}{3} \text{ m/s}$
 (c) $\frac{40}{3} \text{ m/s}$
 (d) $\frac{80}{3} \text{ m/s}$

32. Two second after projection, a projectile is travelling in a direction inclined at 30° to be horizontal. After 1 more second it is travelling horizontally. Then (Take, $g = 10 \text{ ms}^{-2}$)

- (a) the velocity of projection is $20\sqrt{3} \text{ m/s}$
 (b) the angle of projection is 30° with horizontal
 (c) Both (a) and (b) are correct
 (d) Both (a) and (b) are wrong

RESPONSE
GRID

24. (a) (b) (c) (d) 25. (a) (b) (c) (d) 26. (a) (b) (c) (d) 27. (a) (b) (c) (d) 28. (a) (b) (c) (d)
 29. (a) (b) (c) (d) 30. (a) (b) (c) (d) 31. (a) (b) (c) (d) 32. (a) (b) (c) (d)

33. The component of a vector r along x-axis will have maximum value if:

- (a) r is along positive y-axis
- (b) r is along positive x-axis
- (c) r makes an angle of 45° with the x-axis
- (d) r is along negative y-axis

Directions (34-35): These questions consists of two statements each printed as Assertion and Reason. While answering these question you are required to choose any one of the following four responses

- (a) If both Assertion and Reason are correct and Reason is the correct explanation of Assertion.
- (b) If both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.
- (c) If Assertion is true but Reason is false.
- (d) If Assertion is false but Reason is true.

34. Assertion: In case of projectile motion, the magnitude of rate of change of velocity is variable.

Reason: In projectile motion, magnitude of velocity first decreases and then increases during the motion.

35. Assertion: At highest point of a projectile, dot product of velocity and acceleration is zero.

Reason: At highest point, velocity and acceleration are mutually perpendicular.

RESPONSE
GRID

33. (a) (b) (c) (d) **34.** (a) (b) (c) (d) **35.** (a) (b) (c) (d)

Measure Your Performance

Total Questions	35	Total Marks	140
Correct Answers (+4)		Incorrect Answers (-1)	
Total Score			
< 60 Clear Your Chapter Concepts	60 – 100 Revise Important Concepts	100 – 130 Practice More Questions	> 130 Keep Challenging Yourself